DBSCAN Clustering

Density-Based Spatial Clustering of Applications with Noise

Density based clustering Method

- Density based clustering is to detect clusters of non-spherical or arbitrary shapes. Finding high-density regions and outliers.



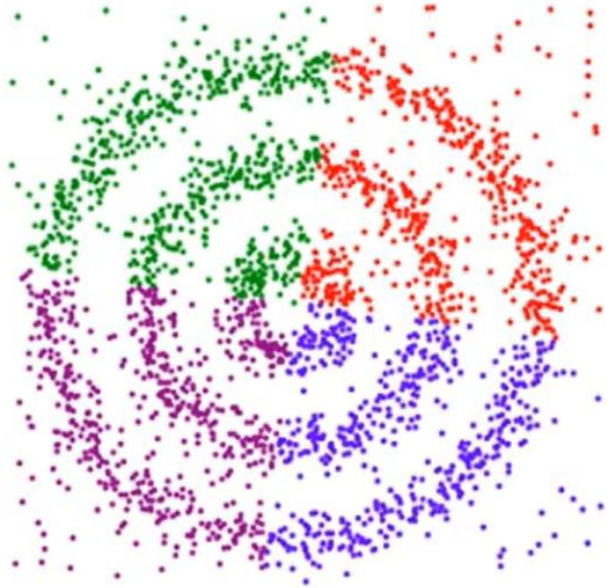
DBSCAN

- **D**ensity-**B**ased **S**patial **C**lustering of **A**pplications with **N**oise (DBSCAN)
- It is able to find arbitrary shaped clusters and clusters with noise (i.e. outliers).
- DBSCAN is a clustering method that is used in machine learning to separate clusters of **high density from clusters of low density.**

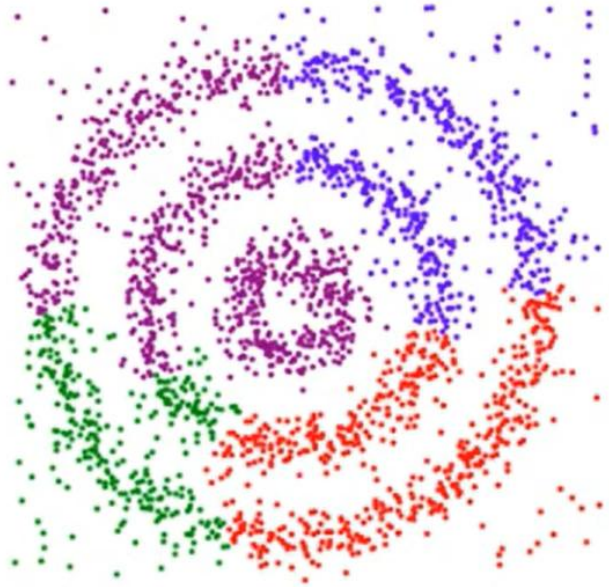
Why need DBSCAN?



DATASET



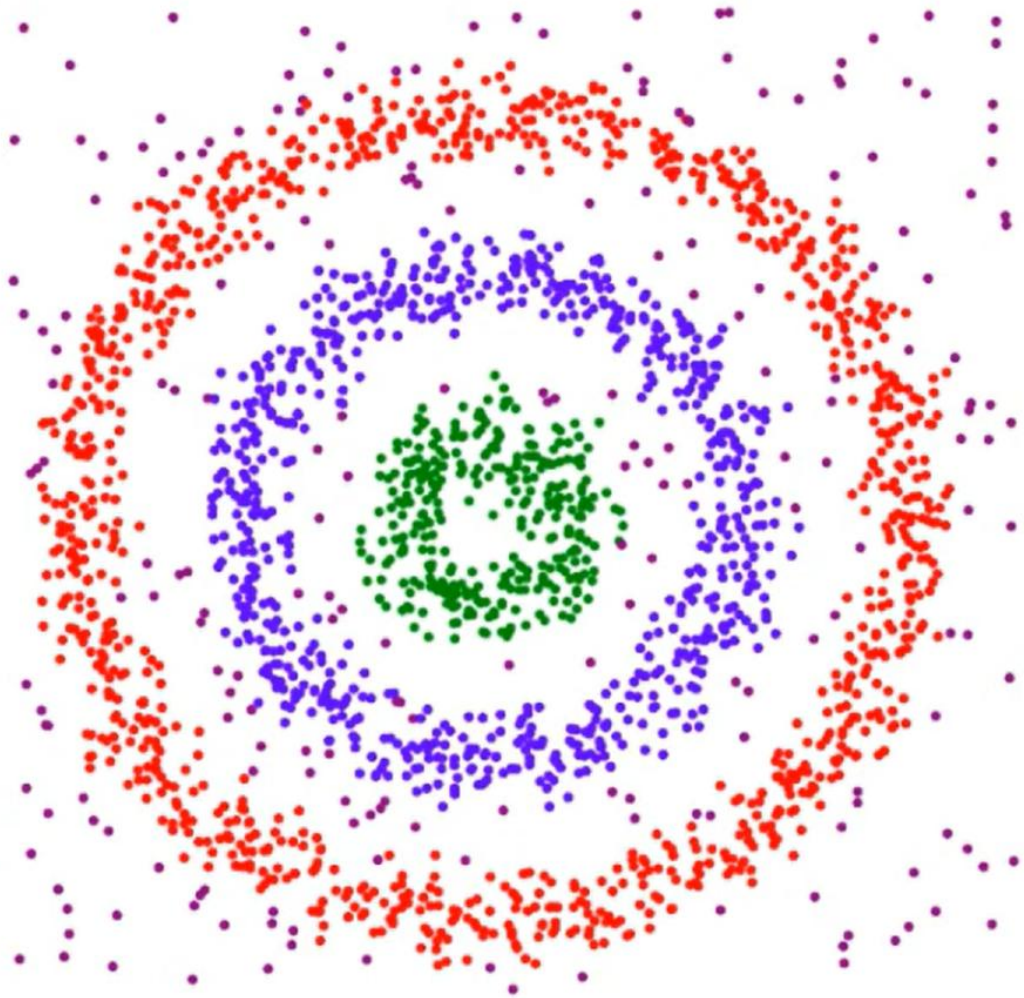
K-MEANS



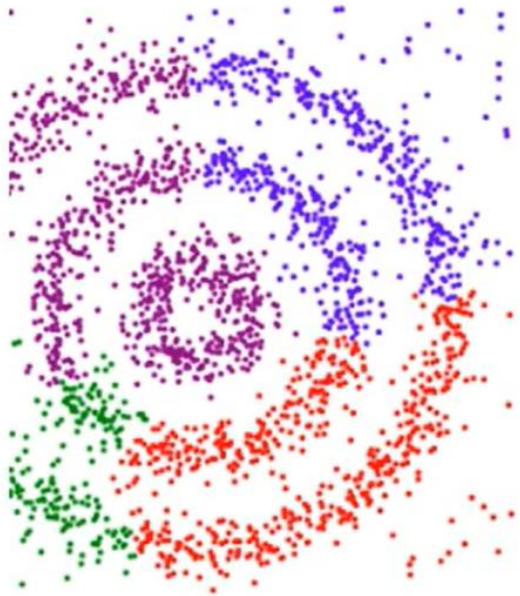
Hierarchical Clustering



DATASET



DBSCAN



Hierarchical Clustering

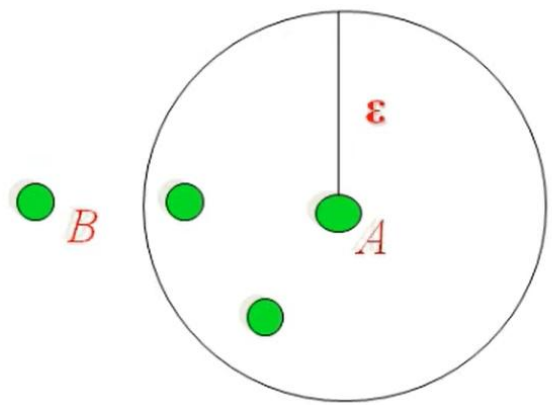
How DBSCAN is Better than K-Means Clustering

Feature	K-Means	DBSCAN
Number of clusters	Must be specified in advance (K)	Found automatically from the data
Cluster shape	Spherical / convex only	Arbitrary shapes (rings, crescents, spirals)
Noise & outliers	Every point is forced into a cluster	Explicitly labels noise points as outliers
Outlier sensitivity	High – outliers drag the centroids	Low – outliers do not affect the clusters
Cluster size / density	Assumes similar-sized clusters	Clusters can be of any size
Initialization	Depends on random centroids; results vary	Deterministic apart from border-point order
Parameters	K	Eps (ε) and MinPts
Best suited for	Compact, well-separated groups	Dense regions separated by sparse areas

Bottom line: DBSCAN handles non-spherical clusters and noise without needing K – exactly where K-Means fails.

DBSCAN requires two parameters:

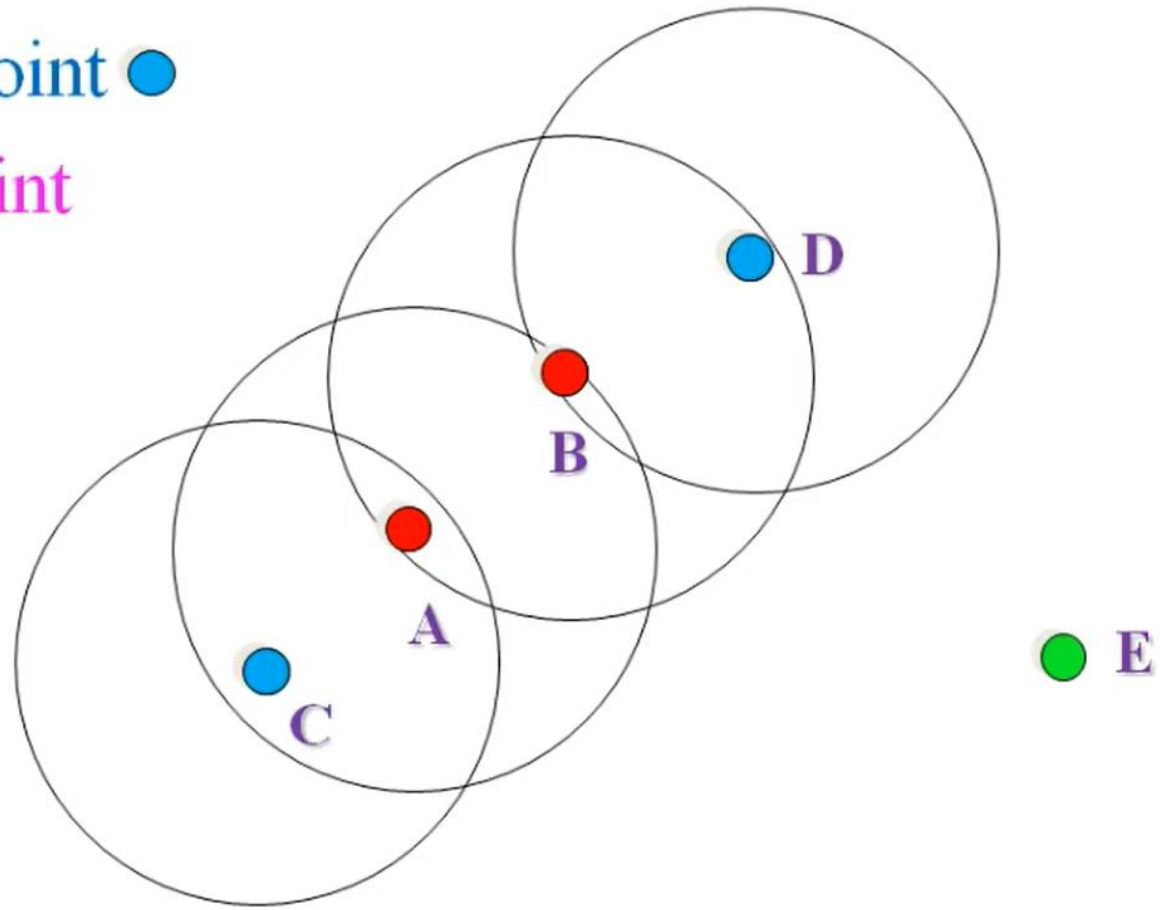
- Epsilon (ε) is the radius of our neighborhoods around a data point. (Circle to be created around each data point)
- minPoints is the minimum number of data points required inside that.



ε= 1cm

DBSCAN categories the data points into three categories:

1. Core point
2. Border point
3. Noise point



ε= 1cm

MinPts = 3

Types of Points

Core point:

A point is a core point if it has more than a specified number of points (MinPts) within Eps.

Border point:

A point is a border point if it is reachable from a core point and there are less than minPts number of points within its surrounding area.

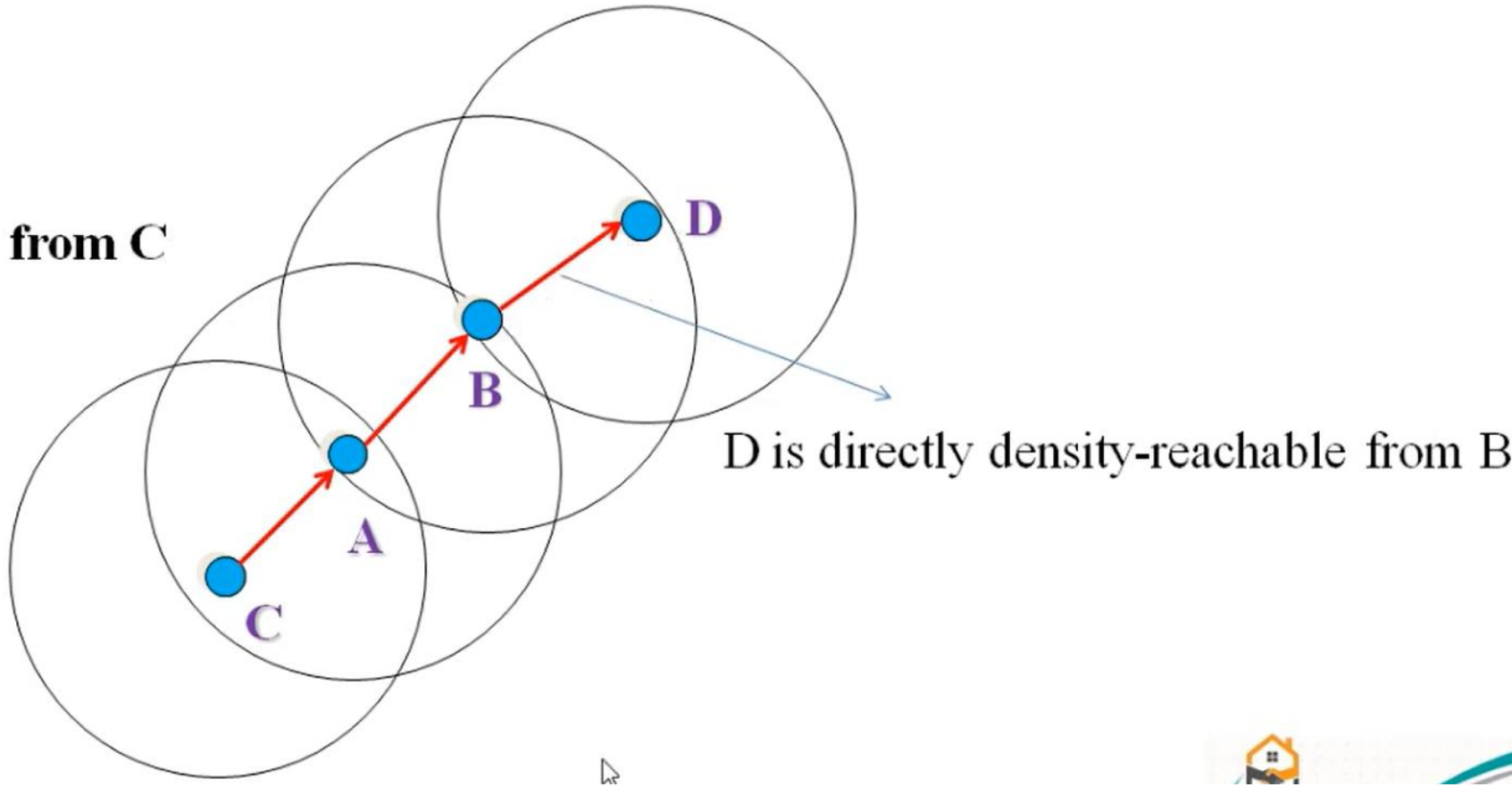
Noise point:

A point is an outlier if it is not a core point and not reachable from any core points.

Density- Reachability

Density-Reachable (directly and indirectly):

D is (indirectly) density-reachable from C



Density Reachability – Definitions

Directly density-reachable

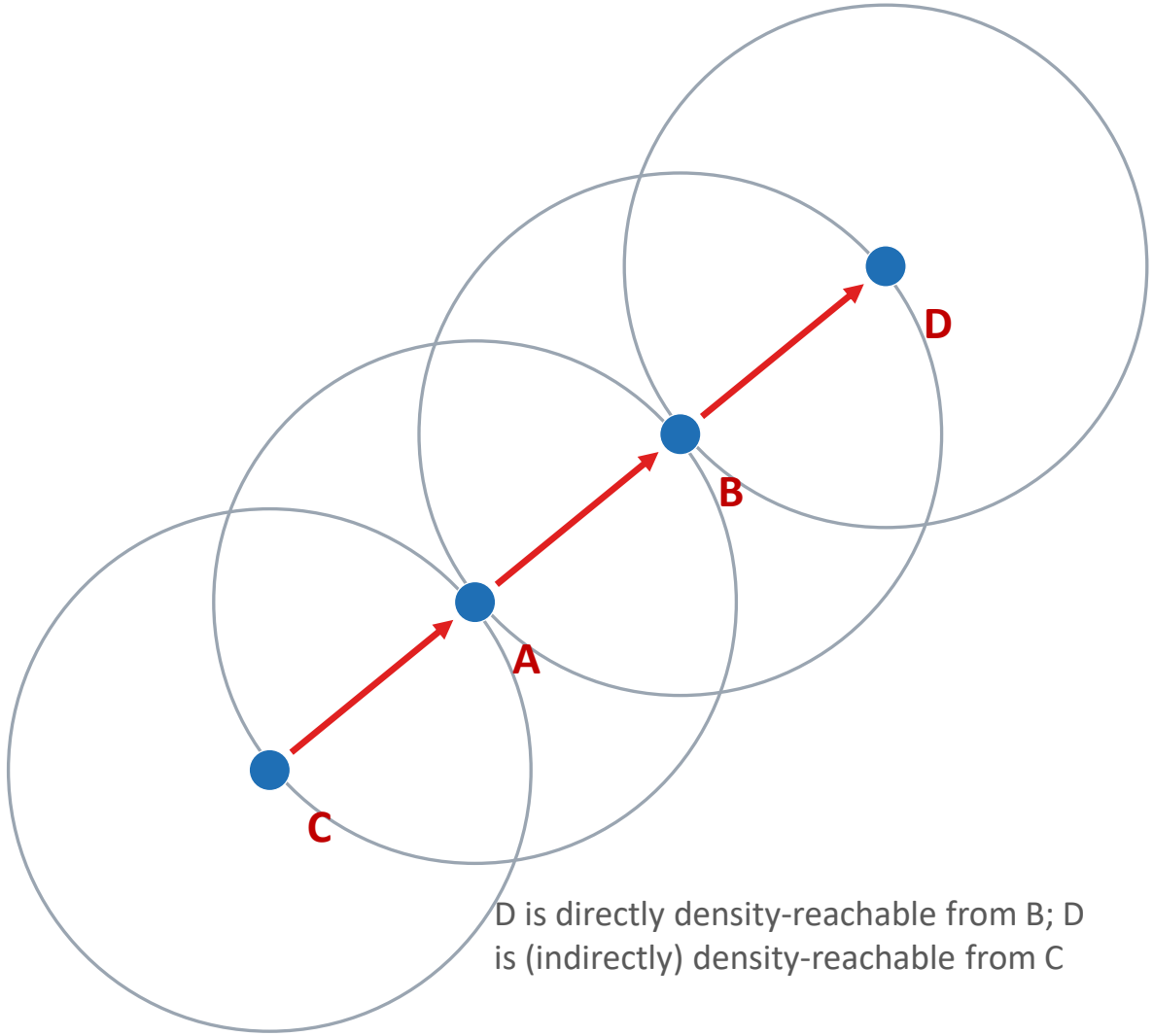
A point q is directly density-reachable from a point p if q lies within ε of p and p is a core point (at least MinPts points within ε).

Density-reachable

A point q is density-reachable from p if there is a chain of points $p = p_1, p_2, \ldots, p_n = q$ such that each p_{i+1} is directly density-reachable from p_i.

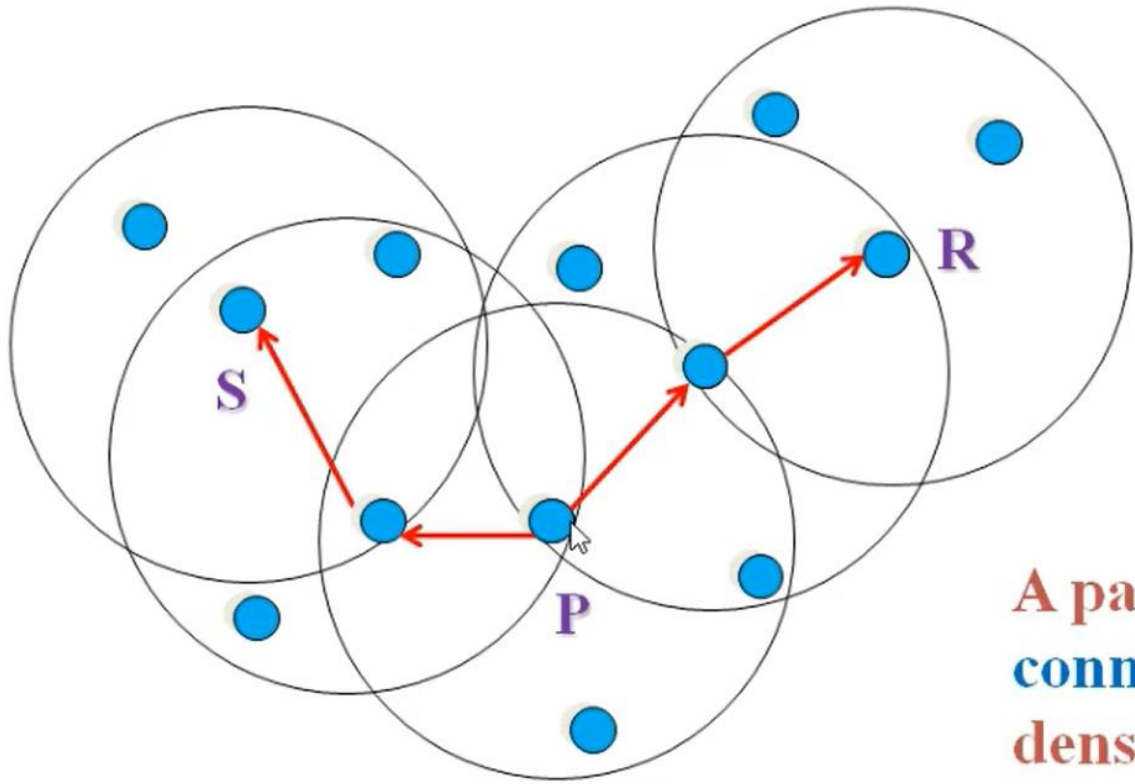
Remember

Density-reachability is not symmetric: a border point can be reached from a core point, but not the other way round.



D is directly density-reachable from B; D is (indirectly) density-reachable from C

Density-Connectivity



A pair of points S and R are **density-connected** if they are commonly density-reachable from a point P.

Density Connectivity – Definition

Density-connected

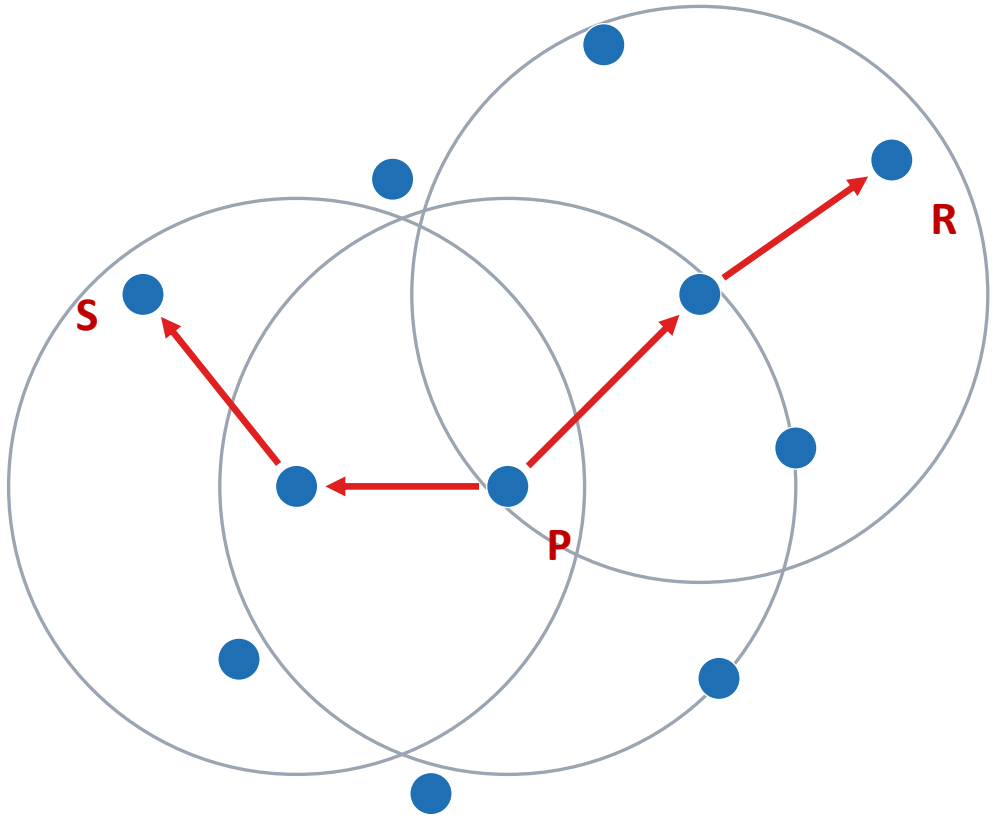
Two points p and q are density-connected if there is a point o such that both p and q are density-reachable from o (with respect to ε and MinPts).

Example

S and R are density-connected because both are density-reachable from the point P.

Why it matters

Density-connectivity is symmetric. A cluster is a maximal set of density-connected points; points connected to no cluster are noise.



S and R are density-connected through P

DBSCAN Algorithm

Input: The data set D

Parameter: ε, MinPts

For each object p in D

    if p is a core object and not processed then

        C = retrieve all objects density-reachable from p

        mark all objects in C as processed

        report C as a cluster

    else mark p as outlier

    end if

End For

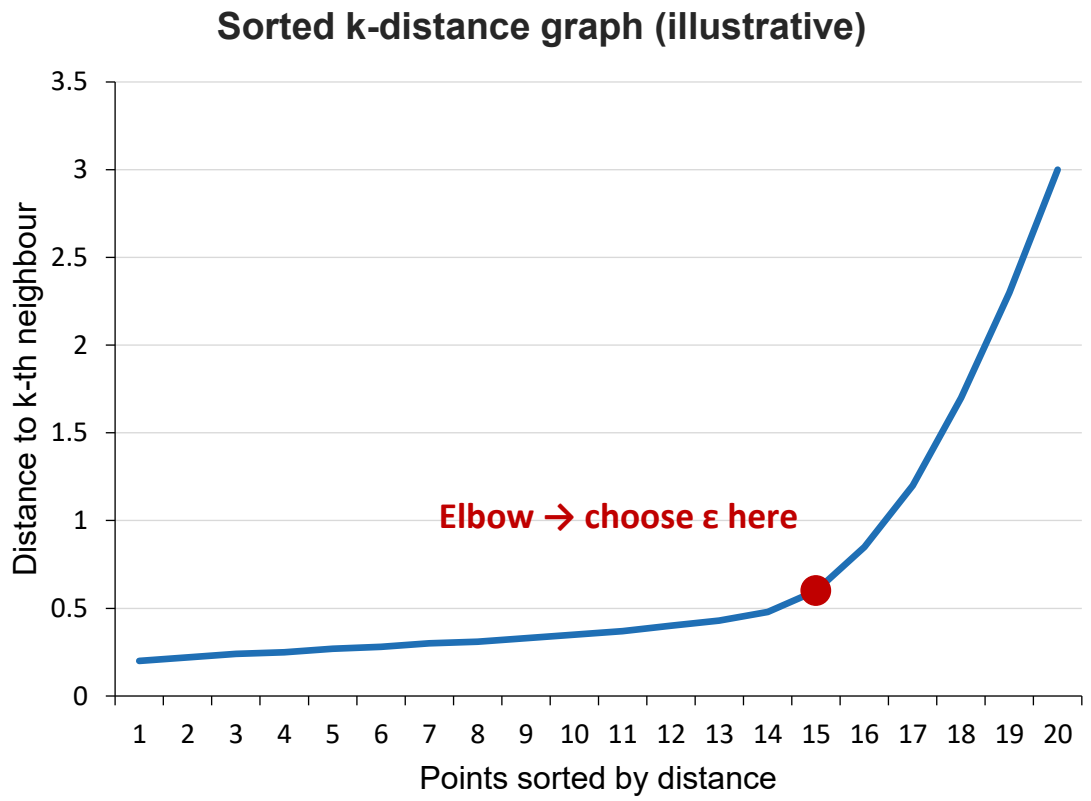
Choosing ε and MinPts

Epsilon (ε) – neighbourhood radius

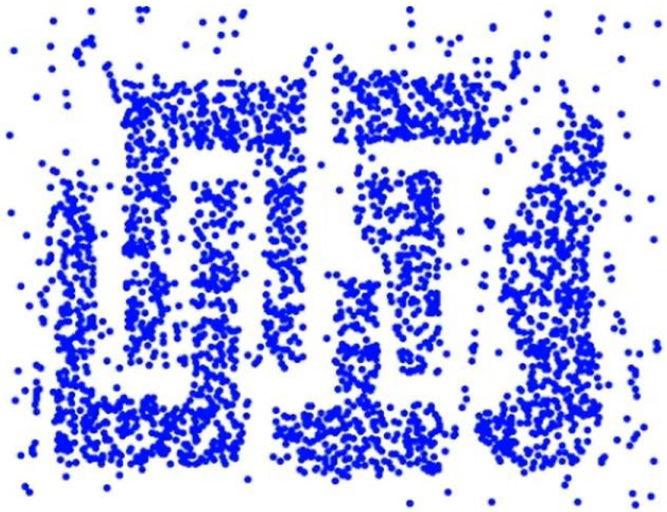
- Plot the sorted k-distance graph (k = MinPts) and pick ε at the “elbow”
- Too small ε → many points become noise
- Too large ε → separate clusters merge into one

MinPts – density threshold

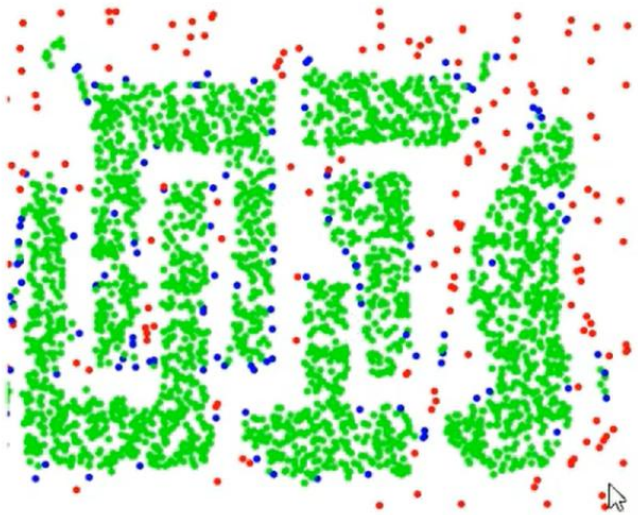
- Rule of thumb: MinPts ≥ dimensions + 1, commonly 2 × dimensions
- Use a larger MinPts for noisy or large datasets
- Features should be scaled before computing distances



EXAMPLE



Original Points



Point types: core, border and outliers

DBSCAN Clustering Algorithm Solved Example – 1

- Apply the DBSCAN algorithm to the given data points and
- Create the clusters with
- **minPts = 4** and
- **epsilon (ε) = 1.9.**

Data Points:

P1: (3, 7)	P2: (4, 6)
P3: (5, 5)	P4: (6, 4)
P5: (7, 3)	P6: (6, 2)
P7: (7, 2)	P8: (8, 4)
P9: (3, 3)	P10: (2, 6)
P11: (3, 5)	P12: (2, 4)

DBSCAN Clustering Algorithm Solved Example

Step 1:

- To find the core points, boundary points and outliers by using DBSCAN we need to first calculate the **distance among all pairs** of given data points.
- Let us use **Euclidean distance** measure for distance calculation.

$$P_1(x_1, y_1) \text{ and } P_2(x_2, y_2)$$

$$\mathbf{dist(p_1, p_2) = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}}$$

DBSCAN Clustering Algorithm Solved Example – 1

- Use Euclidean distance and calculate the distance between each points.

$$\mathbf{Distance}(A(x_1, y_1), B(x_2, y_2)) = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$$

DBSCAN Clustering Algorithm Solved Example – 1

minPts = 4 and epsilon (ε) = 1.9

P1: (3, 7)
P2: (4, 6)
P3: (5, 5)
P4: (6, 4)
P5: (7, 3)
P6: (6, 2)
P7: (7, 2)
P8: (8, 4)
P9: (3, 3)
P10: (2, 6)
P11: (3, 5)
P12: (2, 4)

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
P1	0											
P2	1.41	0										
P3	2.83	1.41	0									
P4	4.24	2.83	1.41	0								
P5	5.66	4.24	2.83	1.41	0							
P6	5.83	4.47	3.16	2.00	1.41	0						
P7	6.40	5.00	3.61	2.24	1.00	1.00	0					
P8	5.83	4.47	3.16	2.00	1.41	2.83	2.24	0				
P9	4.00	3.16	2.83	3.16	4.00	3.16	4.12	5.10	0			
P10	1.41	2.00	3.16	4.47	5.83	5.66	6.40	6.32	3.16	0		
P11	2.00	1.41	2.00	3.16	4.47	4.24	5.00	5.10	2.00	1.41	0	
P12	3.16	2.83	3.16	4.00	5.10	4.47	5.39	6.00	1.41	2.00	1.41	0

DBSCAN Clustering Algorithm Solved Example – 1

minPts = 4 and epsilon (ε) = 1.9

P1: P2, P10
P2: P1, P3, P11
P3: P2, P4
P4: P3, P5
P5: P4, P6, P7, P8
P6: P5, P7
P7: P5, P6
P8: P5
P9: P12
P10: P1, P11
P11: P2, P10, P12
P12: P9, P11

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
P1	0											
P2	**1.41**	0										
P3	2.83	**1.41**	0									
P4	4.24	2.83	**1.41**	0								
P5	5.66	4.24	2.83	**1.41**	0							
P6	5.83	4.47	3.16	2.00	**1.41**	0						
P7	6.40	5.00	3.61	2.24	**1.00**	**1.00**	0					
P8	5.83	4.47	3.16	2.00	**1.41**	2.83	2.24	0				
P9	4.00	3.16	2.83	3.16	4.00	3.16	4.12	5.10	0			
P10	**1.41**	2.00	3.16	4.47	5.83	5.66	6.40	6.32	3.16	0		
P11	2.00	**1.41**	2.00	3.16	4.47	4.24	5.00	5.10	2.00	**1.41**	0	
P12	3.16	2.83	3.16	4.00	5.10	4.47	5.39	6.00	**1.41**	2.00	**1.41**	0

Highlighted: distance ≤ ε (1.9) → neighbours

DBSCAN Clustering Algorithm Solved Example – 1

minPts = 4 and epsilon (ε) = 1.9

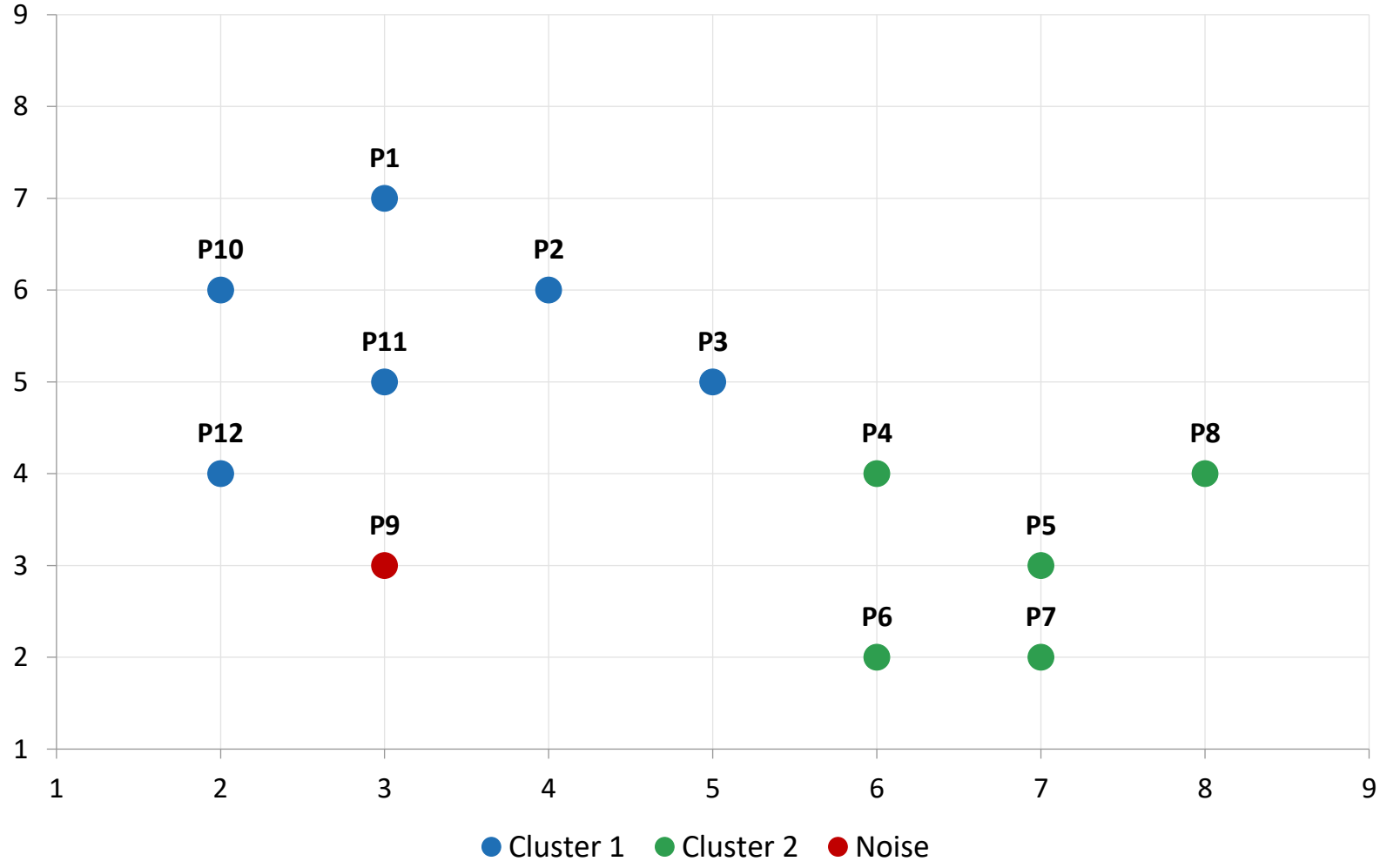
P1: P2, P10
P2: P1, P3, P11
P3: P2, P4
P4: P3, P5
P5: P4, P6, P7, P8
P6: P5, P7
P7: P5, P6
P8: P5
P9: P12
P10: P1, P11
P11: P2, P10, P12
P12: P9, P11

Point	Status	
P1	Noise	Border
P2	**Core**	
P3	Noise	Border
P4	Noise	Border
P5	**Core**	
P6	Noise	Border
P7	Noise	Border
P8	Noise	Border
P9	**Noise**	
P10	Noise	Border
P11	**Core**	
P12	Noise	Border

DBSCAN Clustering Algorithm Solved Example – 1

minPts = 4 and epsilon (ε) = 1.9

P1: P2, P10
P2: P1, P3, P11
P3: P2, P4
P4: P3, P5
P5: P4, P6, P7, P8
P6: P5, P7
P7: P5, P6
P8: P5
P9: P12
P10: P1, P11
P11: P2, P10, P12
P12: P9, P11



DBSCAN Clustering Algorithm Solved Example

- Given the points
- **A(3, 7), B(4, 6), C(5, 5), D(6, 4), E(7, 3), F(6, 2), G(7, 2) and H(8, 4)**
- Find the core points, boundary points and outliers using DBSCAN.

DBSCAN Clustering Algorithm Solved Example

- Given,
- **Epsilon (Eps) = 2.5**
- **Minimum Points (MinPts) = 3**
- Let's represent the given data points in tabular form:

Data Points	X	Y
A	3	7
B	4	6
C	5	5
D	6	4
E	7	3
F	6	2
G	7	2
H	8	4

DBSCAN Clustering Algorithm Solved Example

In the table, Distance ≤ Epsilon (i.e. 2.5) is marked **red**.

Data Points	A	B	C	D	E	F	G	H
A	0	**1.4142**	2.8284	4.2426	5.6569	5.8310	6.4031	5.8310
B		0	**1.4142**	2.8284	4.2426	4.4721	5	4.4721
C			0	**1.4142**	2.8284	3.1623	3.6056	3.1623
D				0	**1.4142**	**2**	**2.2361**	**2**
E					0	**1.4142**	**1**	**1.4142**
F						0	**1**	2.8284
G							0	**2.2361**
H								0

A(3, 7)
B(4, 6)
C(5, 5)
D(6, 4)
E(7, 3)
F(6, 2)
G(7, 2)
H(8, 4)

DBSCAN Clustering Algorithm Solved Example

- Here, data points A, B and C have
- **neighbors < MinPts (i.e. 3)**
- So can’t be considered as **core points**
- Since they belong to the neighborhood of other data points, hence, there exist **no outliers** in the given data points.
- A, B and C are **non-core points** and **not outliers** hence they are called **boundary points**.

N(A) = {B};
N(B) = {A, C};
N(C) = {B, D};
N(D) = {C, E, F, G, H};
N(E) = {D, F, G, H};
N(F) = {D, E, G};
N(G) = {D, E, F, H};
N(H) = {D, E, G};

Advantages & Disadvantages

Advantages

1) Does not require a-priori specification of number of clusters.

2) Able to identify noise data while clustering.

3) DBSCAN algorithm is able to find arbitrarily size and arbitrarily shaped clusters.

Disadvantages

1) DBSCAN algorithm fails in case of varying density clusters.

2) Fails in case of neck type of dataset.

Applications of DBSCAN

1

Anomaly & fraud detection

Noise points flag unusual transactions or behaviour

2

Geospatial analysis

Find hotspots from GPS, crime or accident locations

3

Image processing

Segment regions and detect objects of arbitrary shape

4

Customer segmentation

Group buyers by density of behaviour, spot outlier customers

5

Sensor & network data

Detect faults and intrusions in streaming data

6

Science & bioinformatics

Cluster galaxies, gene expression and cell data

Key Takeaways

1	**Density-based**	Clusters are dense regions of points separated by sparse regions.
2	**Three point types**	Core, border and noise points – defined by ε and MinPts.
3	**No K needed**	The number of clusters comes from the data, not from the user.
4	**Robust to outliers**	Noise is labelled instead of being forced into a cluster.
5	**Limitations**	Struggles with varying densities and high-dimensional data; sensitive to ε.